

PLANET-NL

Netherlands (**NL**) **P**latform for
Large-scale **A**nalysis of **NET**works

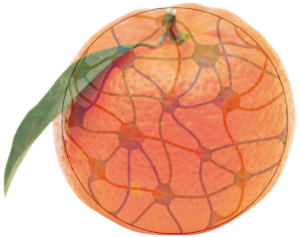
How to model urban segregation

as a *multiscale, multiplex, spatial and temporal* phenomenon

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Yuliia Kazmina
TU Delft - SEGUE / Rbanism Workshop 2026: Segregation computation in R

Universiteit van Amsterdam · Universiteit Leiden · NWO · SSHOC NL



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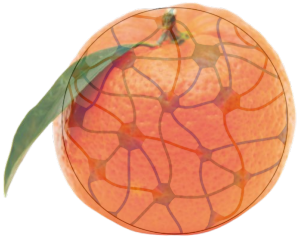
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How to model urban segregation

as a *multiscale, multiplex, spatial and temporal* phenomenon:

A complementary **population-scale social networks** perspective

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Building upon previous points

J. Vallée:



Segregation is a spatial and **temporal** phenomenon.

Residential location alone cannot represent the environments people experience during everyday activities. Need to enrich with mobility trajectories (e.g., occupational segregation).

C. Cottineau-Mugadza and colleagues (SEGUE):

Segregation is produced and reproduced through multiple interacting mechanisms:

economic inequality, housing markets, residential decisions, institutional arrangements, and intergenerational processes.



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I would like to bring forward:

- Unified (reproducible, relatively accessible - to researchers) framework to study multiplex (multi-domain, multi-contextual) segregation
- Segregation **as a property of social structures embedded in space** rather than geographical distributions of population subgroups

Complementary perspectives

Residential

Where people live



Complementary perspectives

School

Occupational

Activity-spaces...



Complementary perspectives

Spatially embedded, affiliation-based

Where people do things



Complementary perspectives

Spatially embedded, affiliation-based

Where people do things



Experienced

Where people go and whom they encounter



Complementary perspectives

Spatially embedded, affiliation-based

Where people do things



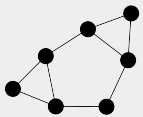
Experienced

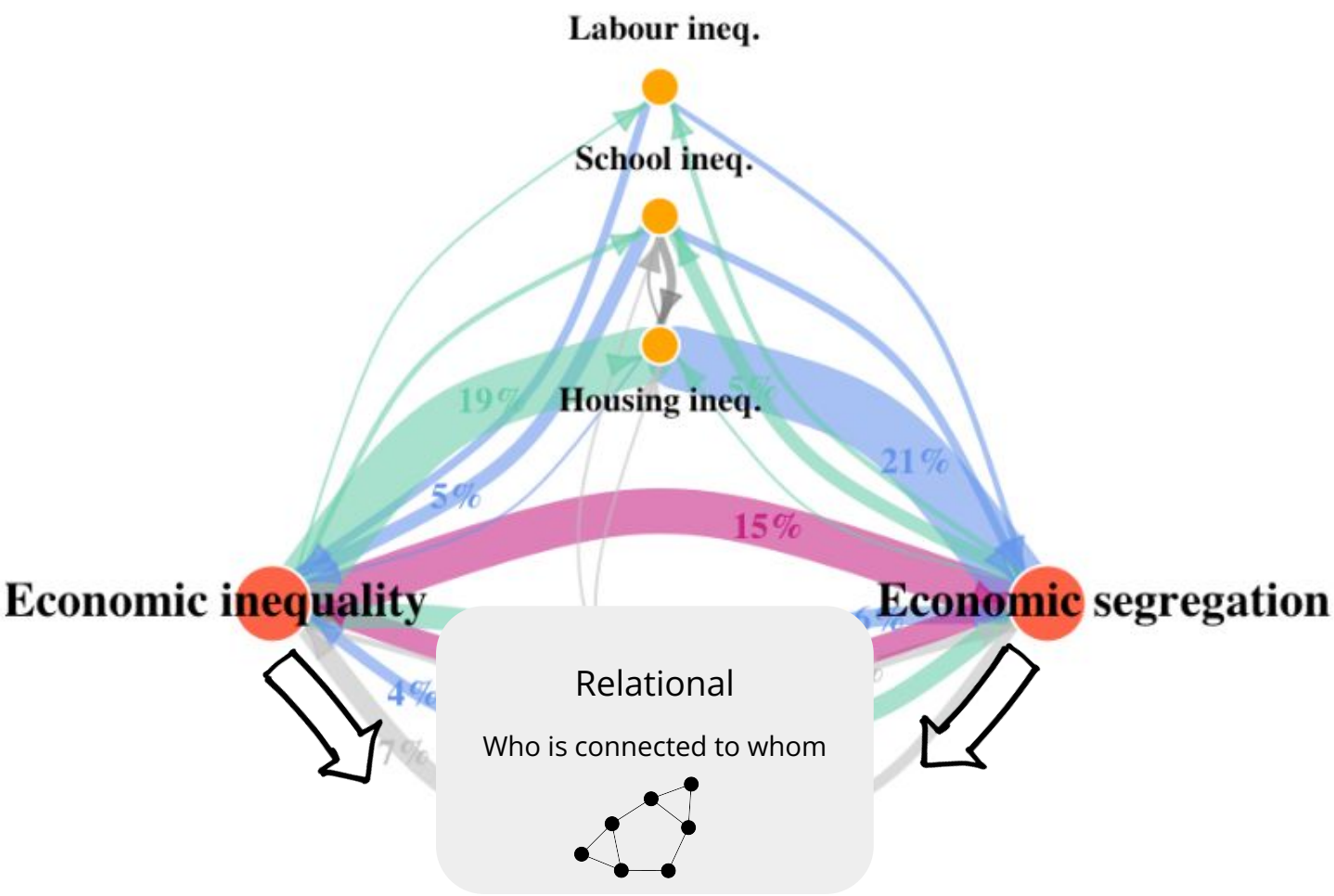
Where people go and whom they encounter



Relational

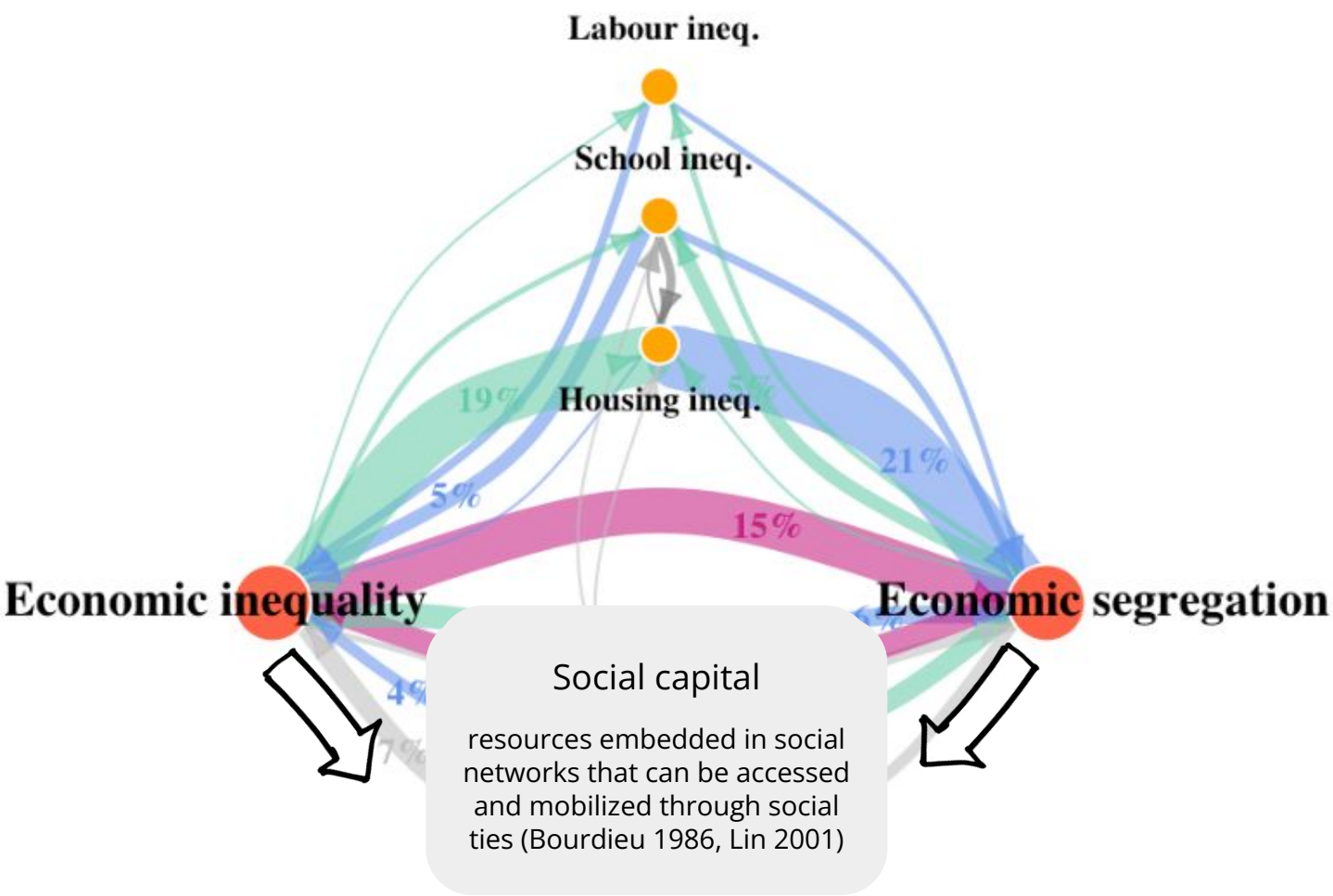
Who is connected to whom





Relational

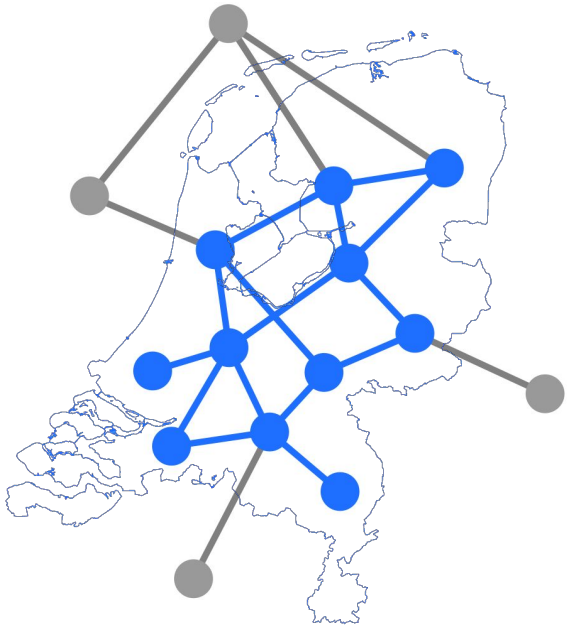
Who is connected to whom



Social capital

resources embedded in social networks that can be accessed and mobilized through social ties (Bourdieu 1986, Lin 2001)

Register-based multilayer population-scale social (?) network

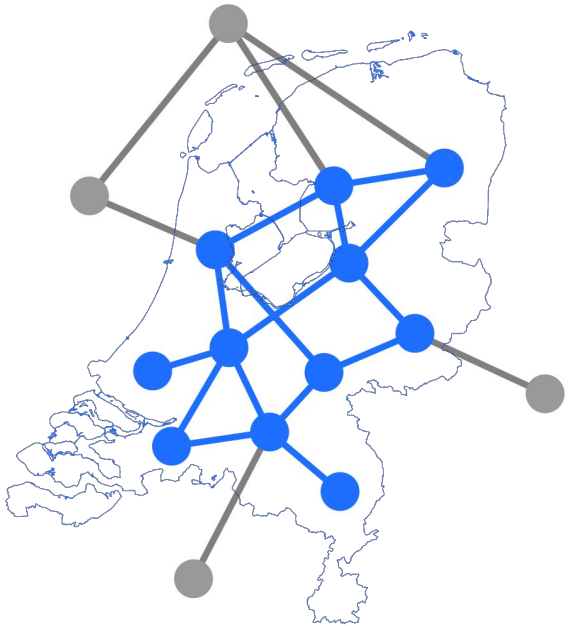


Complete documented population

Register-based relational aspect

van der Laan, J., de Jonge, E., Das, M., Te Riele, S., & Emery, T. (2022). A Whole Population Network and Its Application for the Social Sciences. European Sociological Review

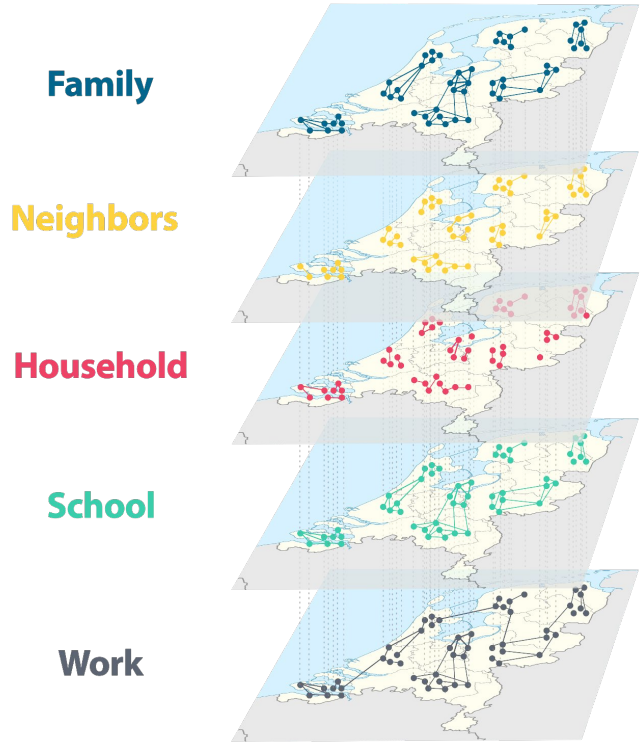
Register-based multilayer population-scale social (?) network



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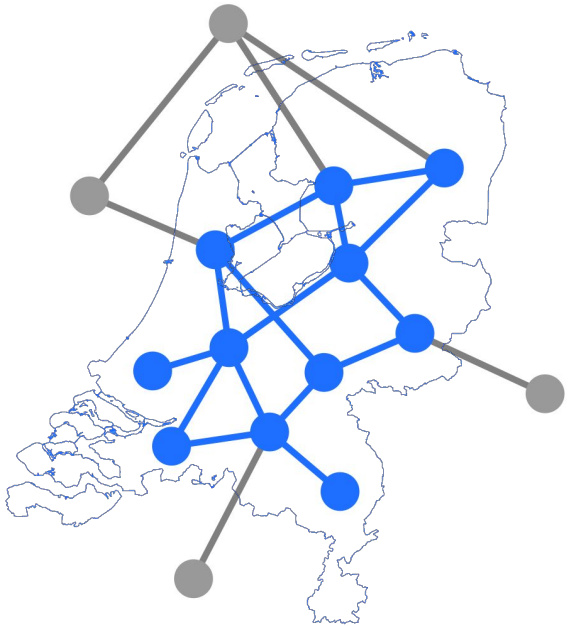
Register-based relational aspect

Multiple domains at once



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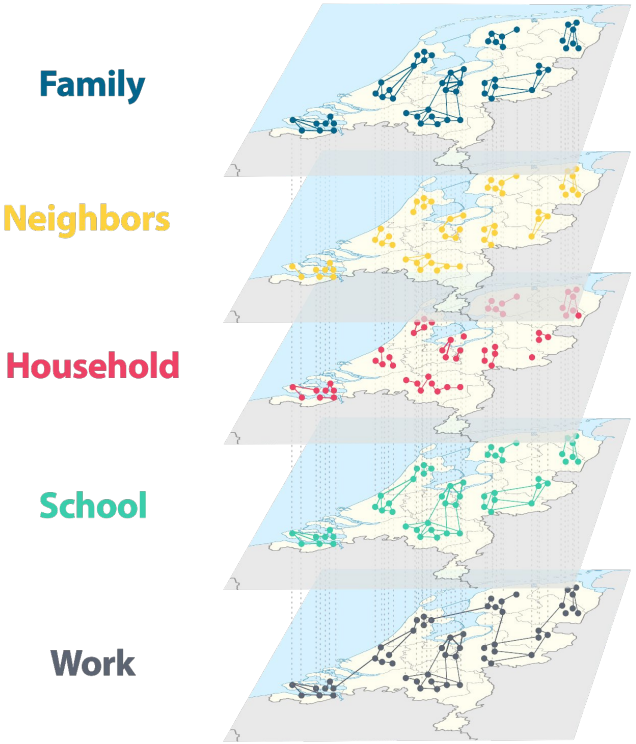


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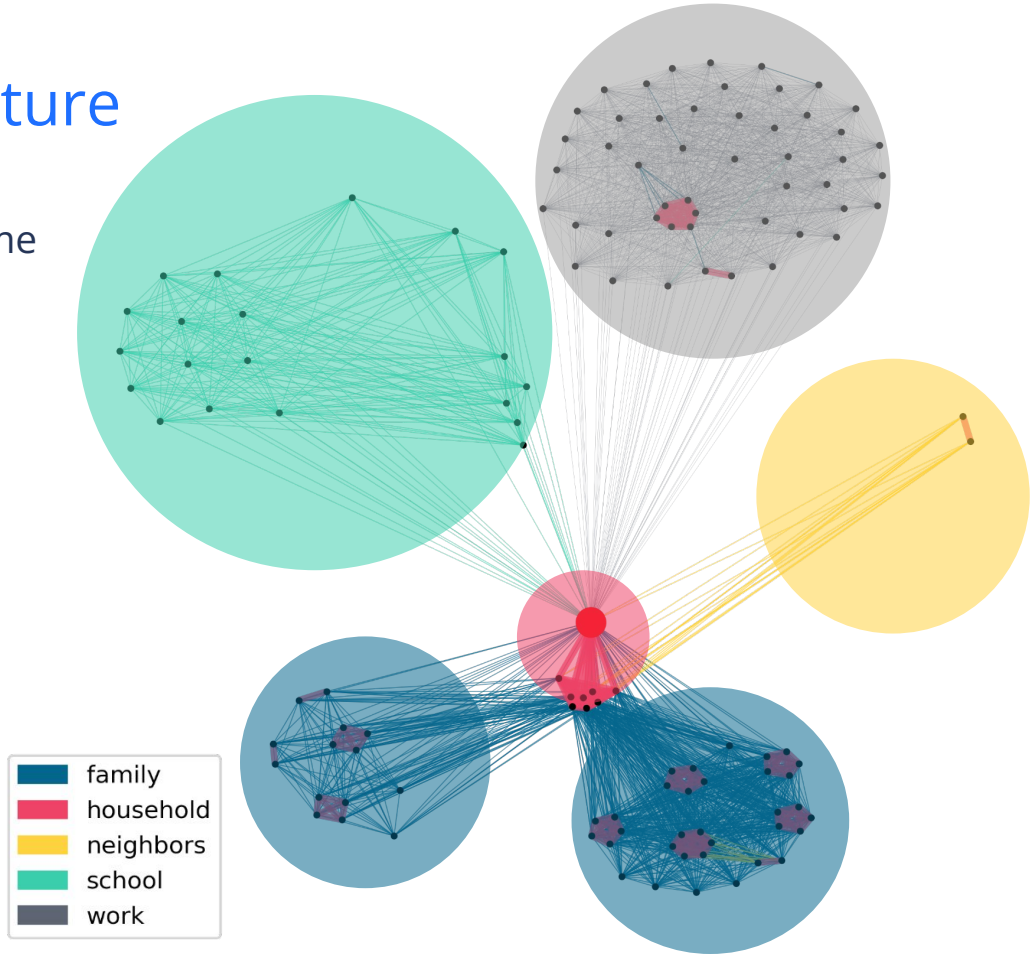
High-quality individual attributes (age, sex, SES, ...)



van der Laan, J., de Jonge, E., Das, M., Te Riele, S., & Emery, T. (2022). A Whole Population Network and Its Application for the Social Sciences. European Sociological Review

Social **opportunity** structure

- Shared institutional settings (some of them are affiliation-like)
- **Structural ties**
- Mix of **strong & weak** ties
- **“Probable friends”** (Dahl & Sorenson, 2010)
- **Latent ties**
- Some of them are **dormant** ties



Feld, S. L. (1981). The focused organization of social ties. American Journal of Sociology, 86(5), 1015–1035. https://doi.org/10.1086/227352
Dahl, M. S., & Sorenson, O. (2010). The social attachment to place. Social Forces, 89(2), 633–658.

Longitudinal and geographic availability



One “snapshot” each year

containing all (documented) residents

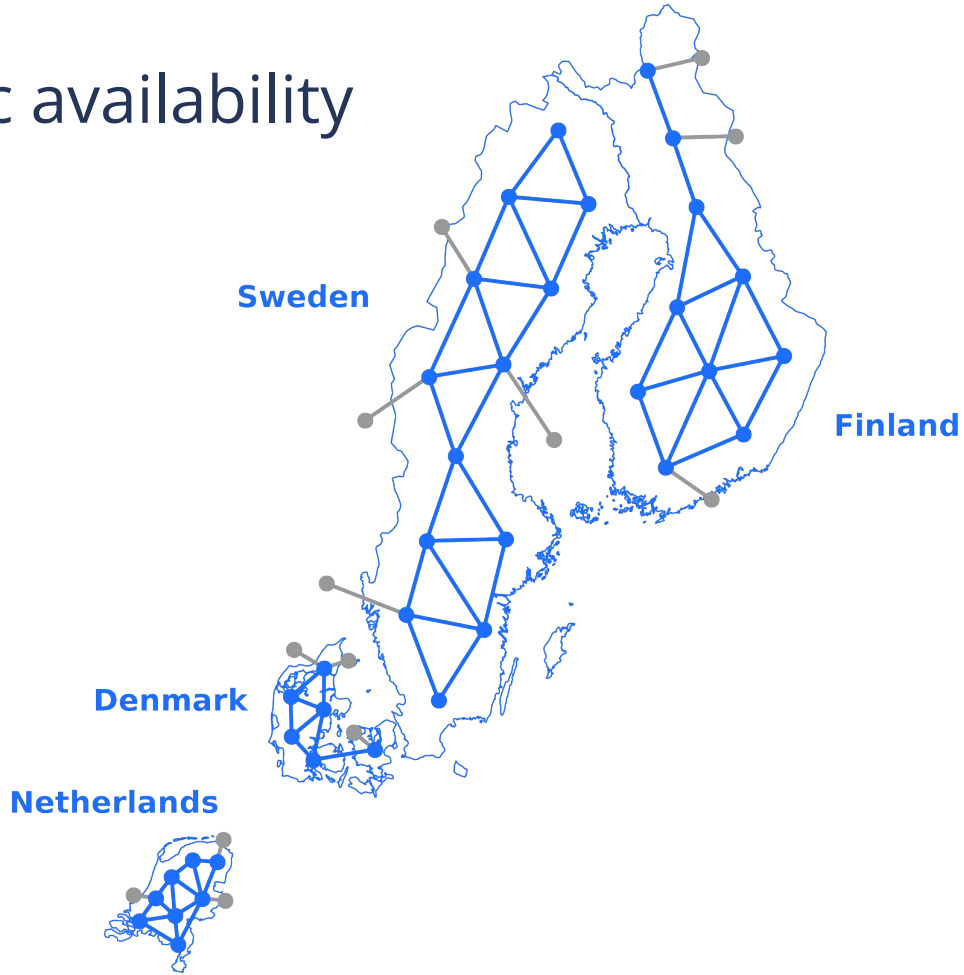
Longitudinal and geographic availability



One “snapshot” each year

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Potential for comparative research



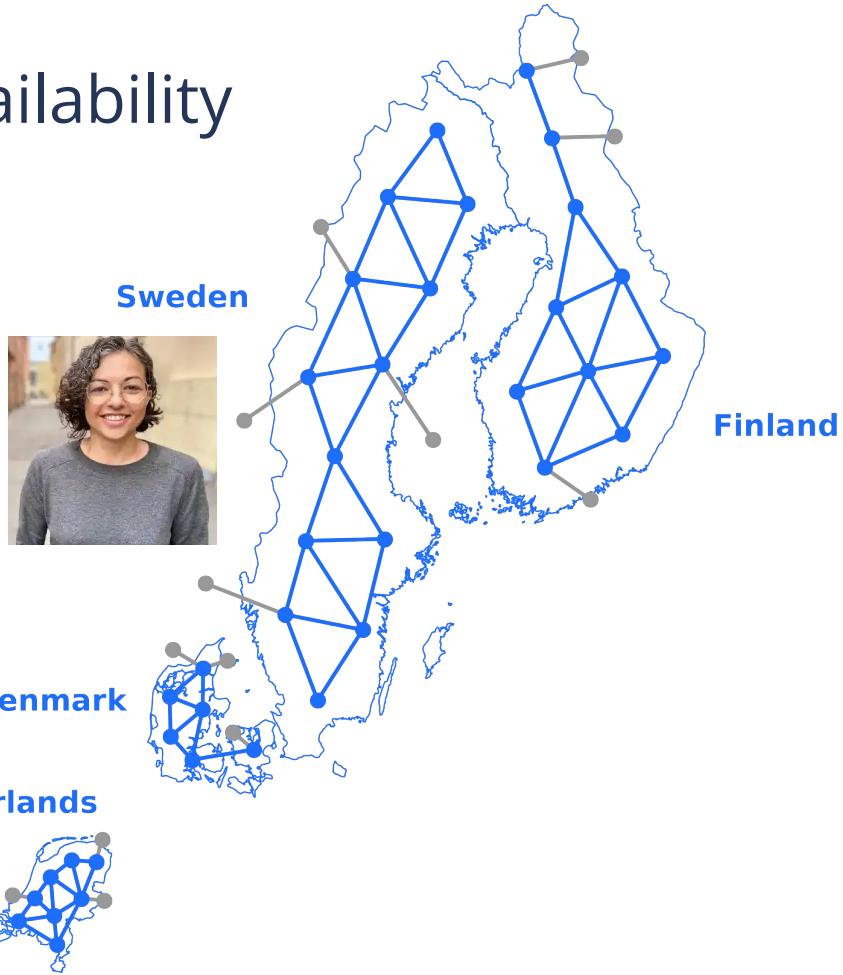
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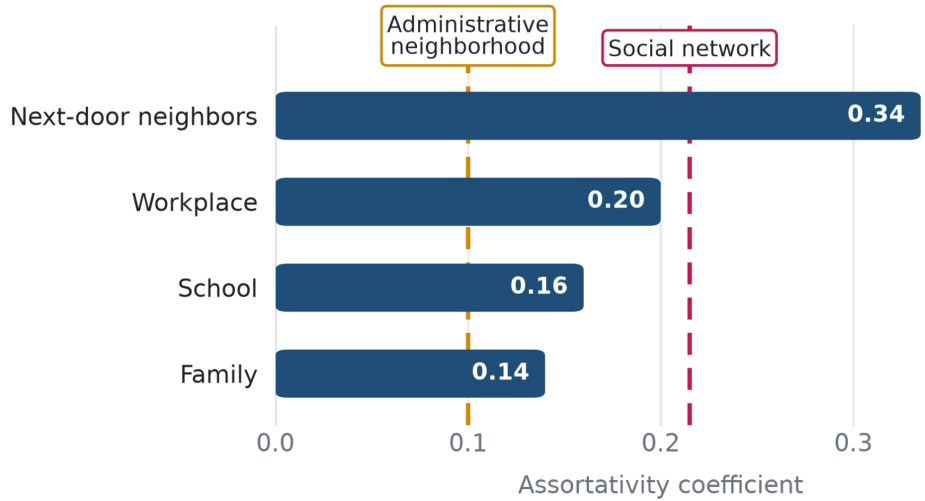
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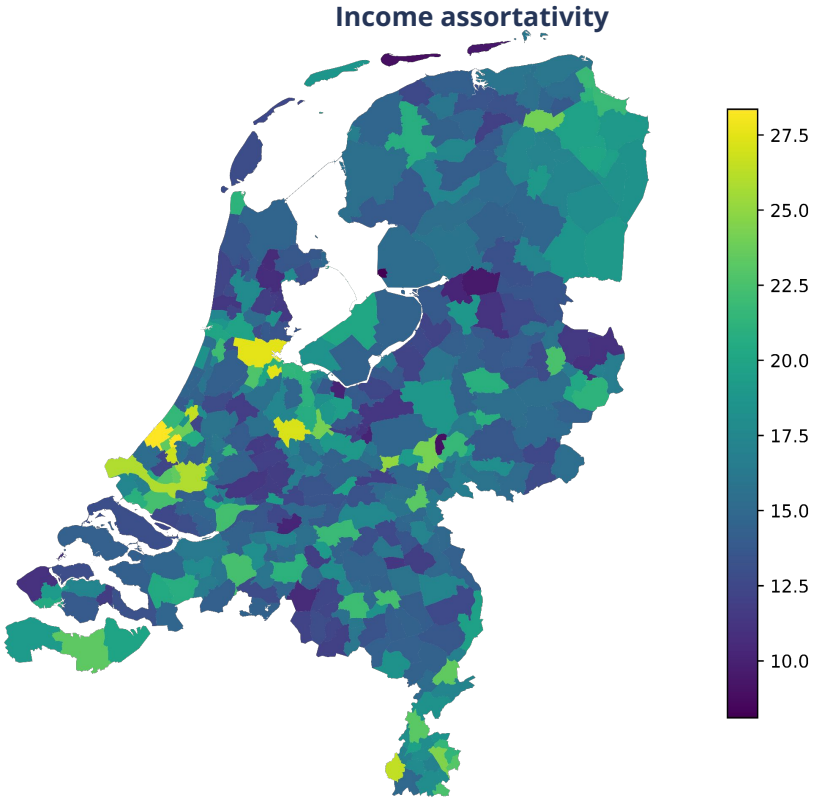
What we learned from this approach... (3)

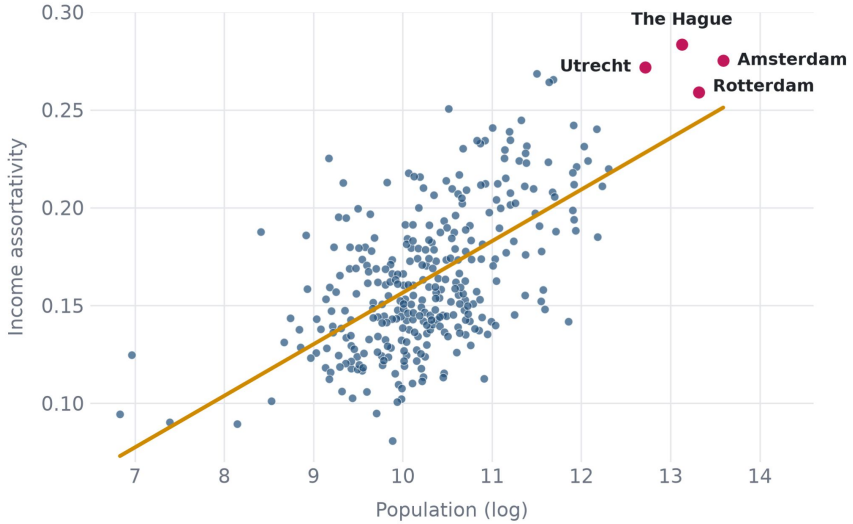
Segregation can remain hidden in mixed neighbourhoods



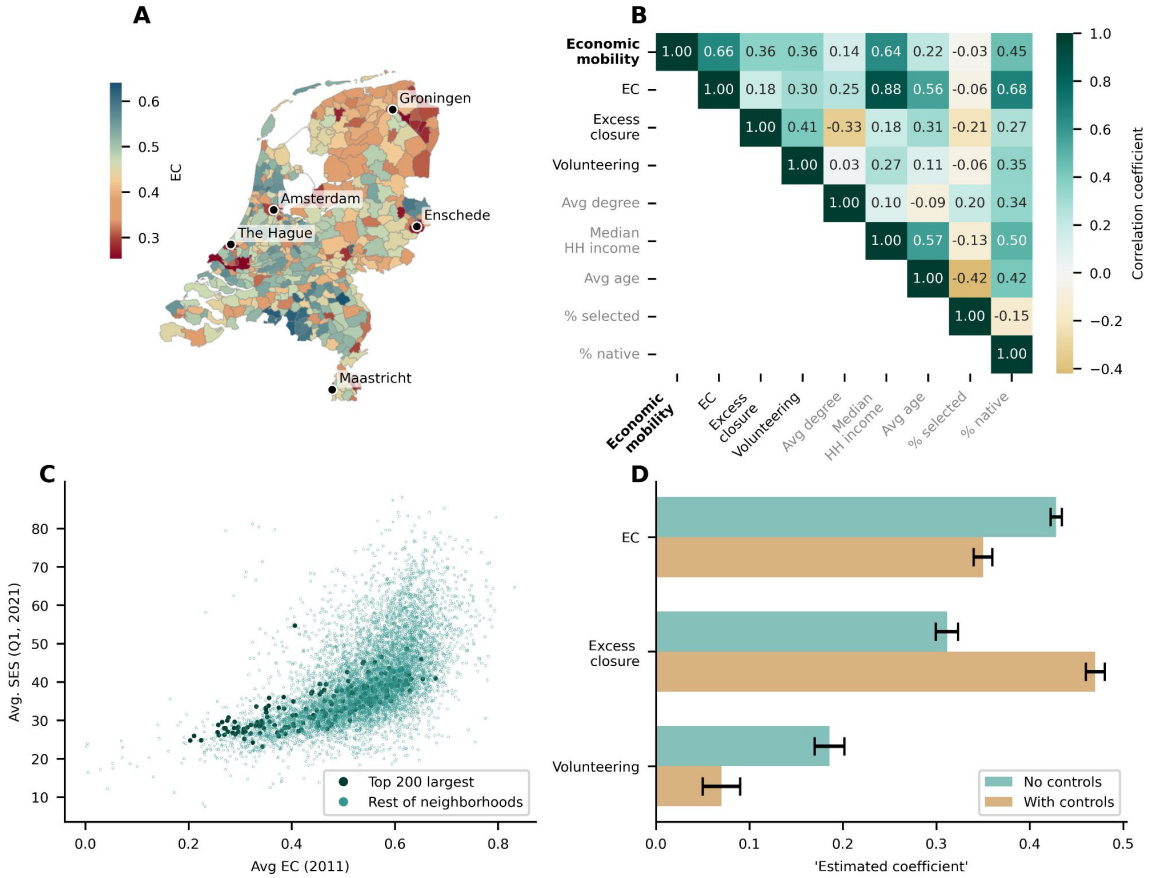
- socioeconomic segregation measured in **social networks** is approximately **twice as high** as segregation measured using administrative neighbourhoods
- each constituent layer of the social network exhibits higher socioeconomic segregation than administrative neighborhoods.
- the highest segregation is observed among **next-door neighbors, followed by workplaces and schools,** while the lowest is observed among family ties.

Larger cities are not the “champions of diversity”

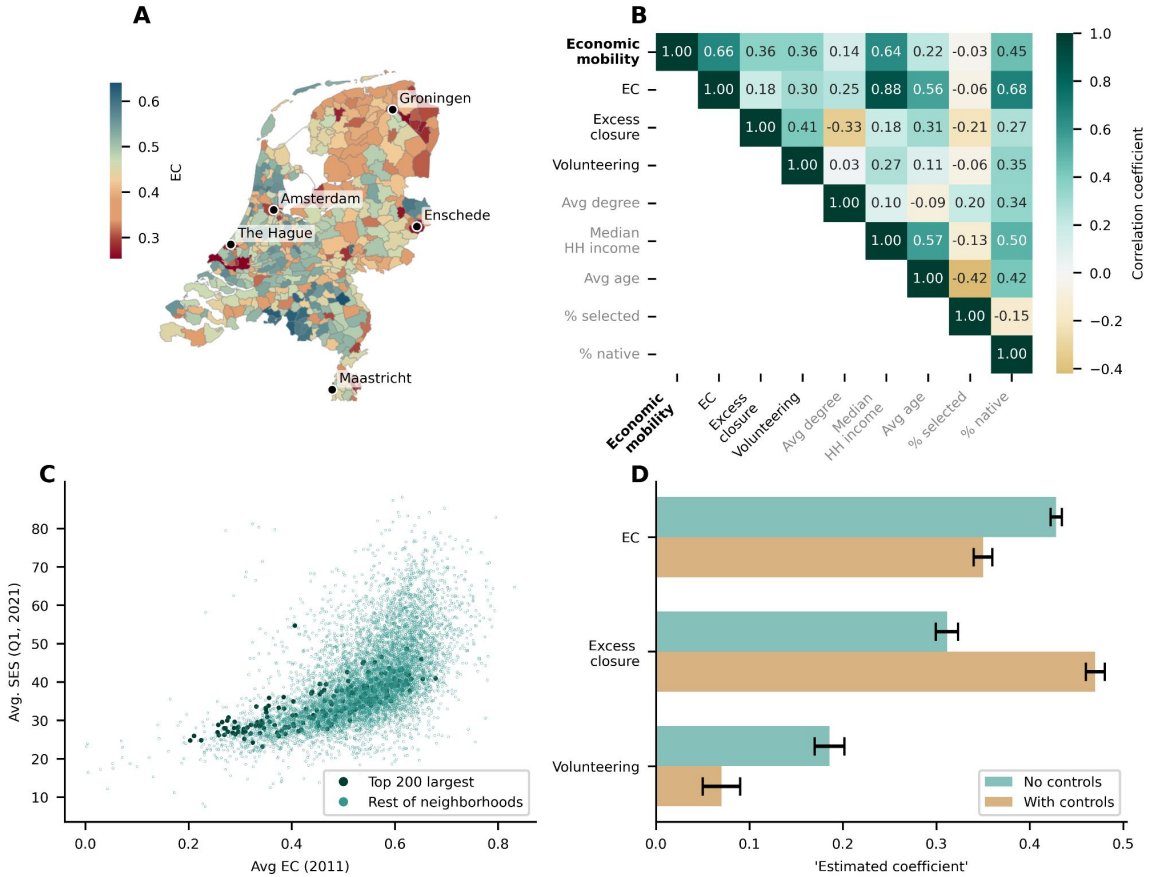




Segregation is both compositional and structural



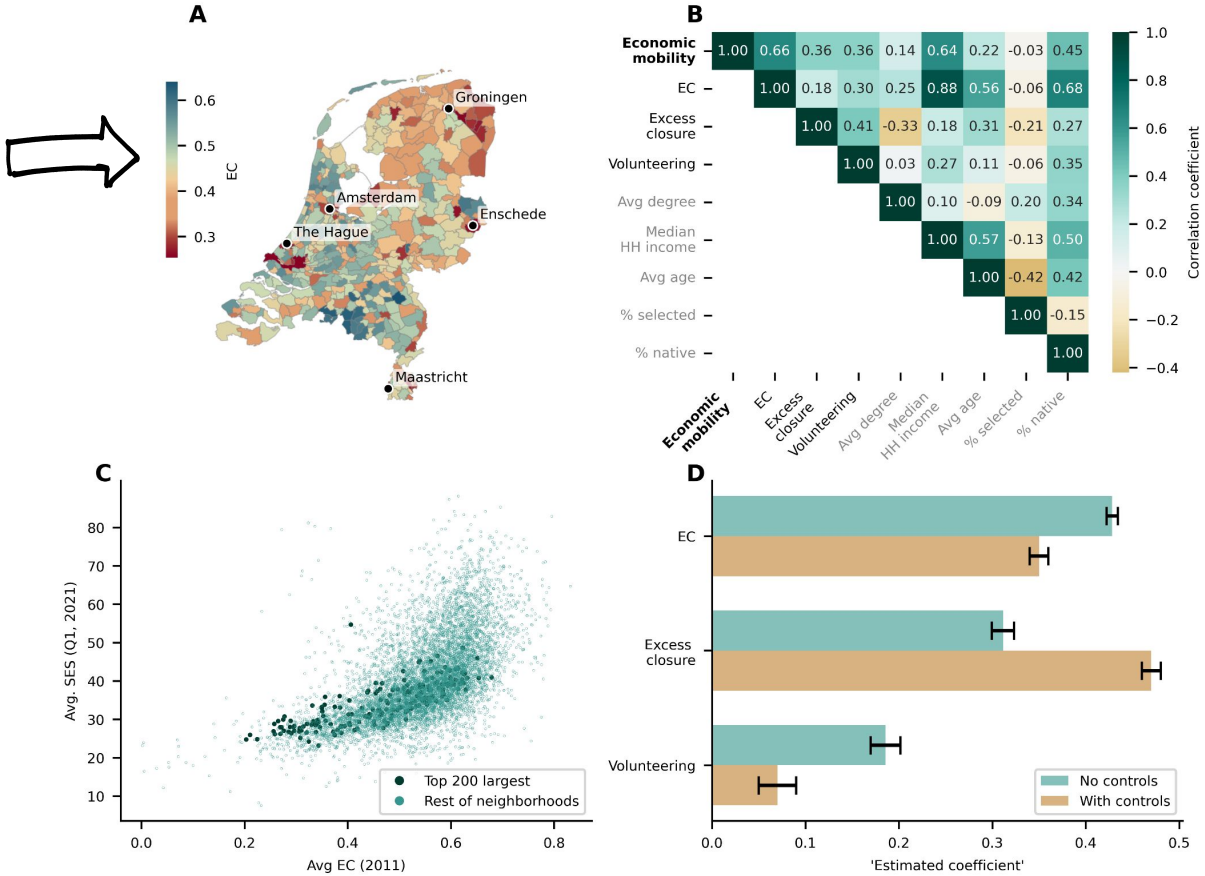
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Compositional:

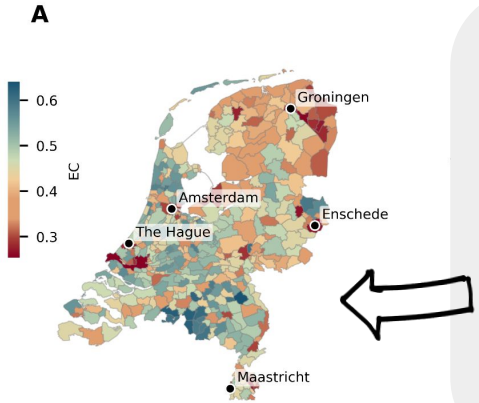
Exposure to affluence among economically disadvantaged groups



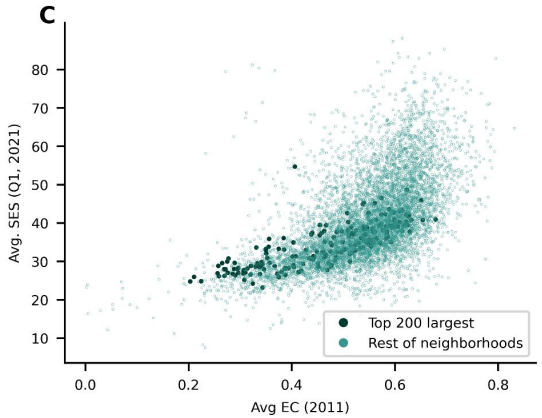
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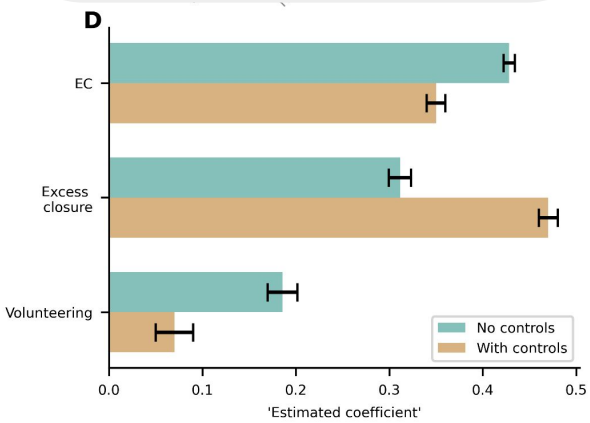
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Greater urban diversity can coexist with greater economic isolation.

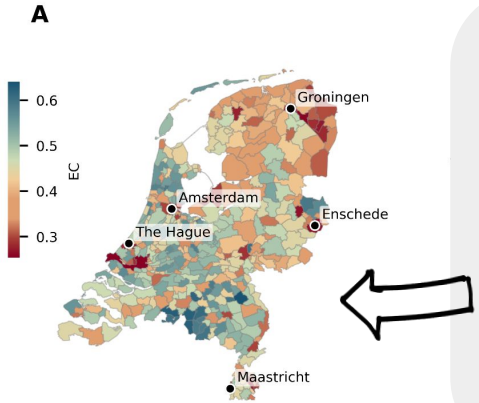




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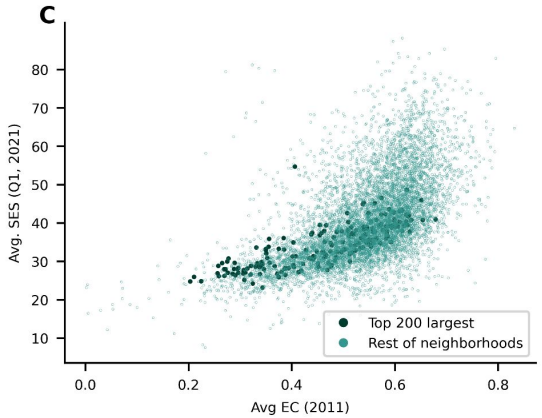
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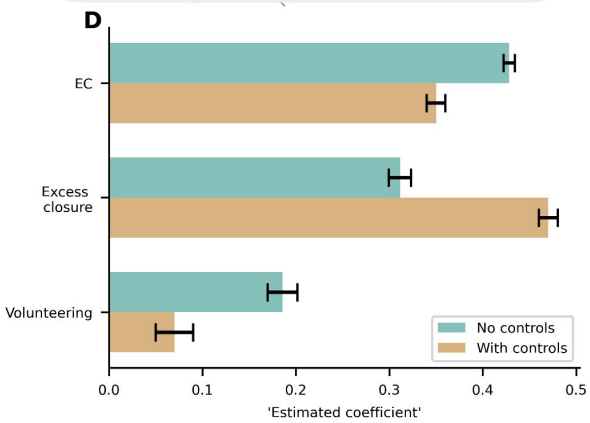
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Exposure to affluence predicting economic mobility

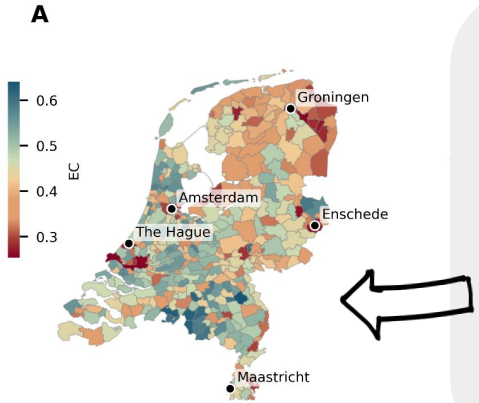




Segregation is both compositional and structural

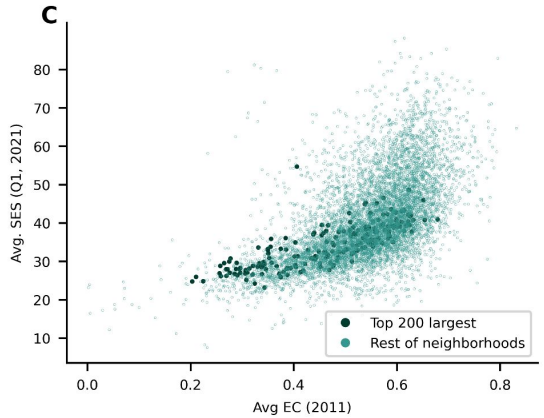
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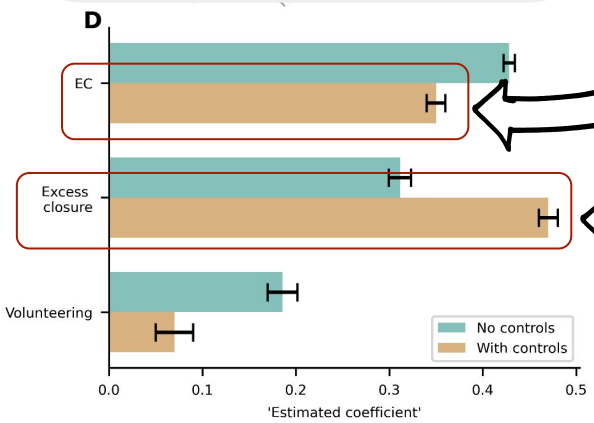
Exposure to affluence among economically disadvantaged groups



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Exposure to affluence predicting economic mobility





Compositional

Structural

Concluding remarks

- Distilling **relational aspect out of the register data** a lot of us are using
- Full-population **multiple-domain** coverage
- Social network can **uncover hidden more pronounced patterns of segregation** that otherwise are not manifesting in spatial analysis
- Enriching compositional segregation with **network-structural**

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